



ULTRON

PERSONAL INTELLIGENCE

User Manual

How to run it, what every screen does,
and how to talk to it

Version 1.0

Frontend v1.0 | Screen vision v1.0

SECTION 1

What Ultron is

Ultron is a personal AI operating layer for your computer: one place to talk to it, and one place it keeps what matters. It holds a conversation, remembers things about you, reads your documents, tracks your tasks and calendar, runs recurring automations, researches topics, and can look at your screen when you ask.

It is built to be **local first**. Your conversation, memories, documents and recordings stay on this machine by default. Cloud models are optional, never required.

This manual covers the current build: how to start it, what each screen does, what you can say, and what is still demo data.

Read this first: what is real today

The interface is complete and every screen works. Behind it, some data is real and some is sample data, because the backend is still being built. Ultron tells you which mode it is in.

Mode	What it means	How you can tell
Demo	No backend is running. Sample projects, tasks and memories are used so you can explore every screen. Changes are saved in your browser only.	A message on start: <i>Demo data - no backend connected.</i>
Live	A real backend answered. Your actual data is used.	No demo message appears.

Ultron checks for a backend automatically on every start, and falls back to demo data rather than showing you a broken screen. Section 8 explains how to connect the real backend.

SECTION 2

Starting Ultron

The interface

Open a terminal in the Ultron folder and run:

```
python frontend\serve.py
```

Then open **http://127.0.0.1:5173** in your browser. That is all - there is nothing to install and nothing to build.

With your backend

```
python frontend\serve.py --api http://127.0.0.1:8000
```

This passes API calls through to your backend so the browser sees a single address. Use it while developing; you will not need to configure anything else.

The desktop app

The Windows app is a separate program in the same project:

```
python -m maira
```

It starts the orb, the tray icon and the global hotkey. It can start automatically when you log in to Windows - see Settings, or the presence section of `config/default.yaml`.

Stopping

Press **Ctrl+C** in the terminal running the server. Nothing is lost: your data is on disk, not in the server.

SECTION 3

The Core: Ultron's presence

The small glowing orb is Ultron itself. It is always somewhere on screen - centred on the home page, in the middle of Focus Mode, and resting in the bottom-right corner everywhere else.

It is not decoration. **The way it moves tells you what Ultron is doing**, so you rarely need to read any text to know its state.

Click the orb to start talking. If it is speaking or listening, clicking again interrupts it - the same as talking over someone.

What each state looks like

State	What it looks like	What it means
Ready	Slow, uneven breathing. Almost still.	Nothing is happening. Ultron is present and waiting.
Listening	Blue, swelling with your voice.	Your microphone is open.
Thinking	Purple light circling inside.	Working out an answer.
Working	Amber, the ring turning steadily.	Running a task, tool, or automation.
Speaking	Teal, pulsing in time with speech.	Replying to you.
Recording	A small red dot on the rim.	Screen recording is on. Always visible - it cannot be hidden.
Approval	Amber, holding still, waiting.	Ultron needs your permission. Answer with the tick or cross.
Done	One soft green bloom, then calm.	The task finished.
Error	Dim red, settling and still.	Something failed. The message says what.

A short label appears beside the orb when something is happening - *Thinking...*, *Running tests...* It says what Ultron is doing, never what it is saying: the reply itself belongs in the conversation.

SECTION 4

Finding your way around

The command bar

The wide field on the home page - *What are you thinking about?* - is the main way in. Type naturally and press **Enter**. **Shift+Enter** makes a new line.

Underneath it are shortcuts for the common things: Ask Ultron, Deep research, Generate ideas, Solve problem, Plan my day, Search memory, Create task. You can also attach files, images or documents with the paperclip, or drag them onto the bar.

The command palette

Press **Ctrl+K** anywhere. Start typing to jump to any screen, run a command, or search across everything at once - conversations, memories, documents, tasks, research, projects and people. Arrow keys to move, Enter to go, Escape to close.

The sidebar

Every section of Ultron, with Focus Mode, Settings and Profile at the bottom. It collapses to icons using the handle on its edge, and does so automatically on a narrow window. On a phone it becomes a row of tabs along the bottom.

Notifications

The bell in the top bar. Ultron interrupts only for things that matter: a reminder falling due, an automation that failed, something important it found. It is not a feed.

Keyboard

Ctrl+K	Open the command palette
/	Open the command palette
Alt+Space	Talk to Ultron
Escape	Close whatever is open
g then o	Go to Overview
g then c	Go to Chat
g then m	Go to Memory
g then t	Go to Tasks
g then k	Go to Knowledge
g then a	Go to Agents
g then r	Go to Research
g then i	Go to Insights
g then s	Go to Settings

SECTION 5

Every screen, and what it is for

Overview

Home. Your greeting, the orb, and the command bar, with a summary underneath: what Ultron did recently, how memory is doing, your focus trend for the week, task progress, and which agents are busy right now. Start here.

Ultron Chat

The main conversation. Replies stream in as they are written. When Ultron uses a tool you can open the step list and see exactly what it did and how long each step took. Where it relied on something it knows, it cites the source.

When a message causes something to happen - a task created, a reminder set, something remembered - a small confirmation card appears under the reply. Click it to go straight there.

Hover any message to copy it, save it to memory, turn it into a task, or ask for a different answer.

Mind

Everything Ultron knows, drawn as a map: you at the centre, then projects, people, knowledge, goals, tasks, memories and concepts. Drag to move, scroll to zoom, click a node for detail, double-click a group to fold it away. Use it when you want to see how things connect rather than read a list.

Memory

What Ultron knows about you, and why. Every memory shows where it came from, how confident it is, how important it is, and when it was last used. You can pin what must never be forgotten, edit anything that is wrong, and delete anything permanently.

Teach Ultron adds something directly. You can also just say it in conversation - *"Remember that I prefer local models"* - and it will be stored.

Knowledge

Your documents: PDFs, notes, links, images and code. Drop a file in and Ultron reads it, summarises it, tags it, and links it to the right project. Open any item to summarise it, ask Ultron about it, or save what it learned to memory.

Tasks

Today, Upcoming, Inbox, Completed and by Project. The important part is the box at the top: write a task the way you would say it.

Remind me every Monday at 9 AM to review the roadmap

As you type, Ultron shows what it understood - the date, the time, whether it repeats, the priority, which project. Press Enter and it is created. A repeating task also becomes an automation.

Plan my day

Ultron reads your calendar, your open tasks and your deadlines, and lays out a realistic day in morning, afternoon and evening - with a reason for each choice, what it suggests postponing, and any conflicts it noticed.

Calendar

Day, week and month. Meetings, deadlines, reminders and focus blocks, colour-coded. Above it, one honest sentence about what is coming: *"You have 3 meetings tomorrow and about 2 free hours."*

Automations

Recurring behaviour. Each one shows its trigger, its action, when it last ran, when it runs next, and how many times it has failed. Pause any of them, run one immediately to test it, or open its full history.

Create one in plain language: *"Every morning summarise my schedule."*

Agents

The specialists Ultron delegates to - research, memory, planning, tasks, knowledge, automation and system control. Each shows what it can do, what it is working on, and its recent runs.

At the top is the orchestration view. Give it something complicated and watch the work pass between agents, step by step, ending in a verified result.

Research

Deep research. Ask a question and Ultron plans, searches, compares, analyses and writes up what it found - live, so you can watch it work. The report gives you a summary, findings, evidence, **contradictions it noticed**, recommendations and next actions. Export it, turn the next actions into tasks, or save it to Knowledge.

Projects

A workspace per project, gathering its tasks, conversations, files, research, memories, goals and activity. While you are inside one, Ultron uses it as context - so *"What's left?"* means that project without you having to say so.

Goals

Bigger objectives with milestones and a deadline. Tick milestones off as you finish them. Ultron adds a recommendation on each - the honest one, not the encouraging one.

Insights

How you actually spent your time: focus, productivity, task completion, weekly work and learning hours, your most active projects, and what you rely on most.

Measured numbers and Ultron's interpretation are always kept separate, and anything interpreted is labelled as such.

Activity

Everything Ultron has done, newest first, grouped by day, filterable by kind. When you want to know what happened and when, look here.

Focus Mode

Everything disappears except one task, a timer, and Ultron. Pick something to work on, start, and close everything else. Pause when you break; press **Escape** to leave - the session keeps running in the background so you can come back to it.

Settings

General, AI, Memory, Voice, Notifications, Privacy and Integrations. Everything saves as you change it. Privacy also holds the data controls - see Section 9.

Profile

Your name, timezone, language, how Ultron should answer you, and which notifications you want.

Search

One search across everything. Reachable from the top bar or with Ctrl+K.

SECTION 6

Talking to Ultron

You do not need commands or special syntax. Say what you want the way you would say it to a person. These are examples, not a list of everything it accepts.

Getting things done

Remind me tomorrow at 9 AM to review the roadmap	Creates a reminder
Remind me every Monday at 9 AM to check the backlog	Creates a repeating reminder and an automation
Create a task to fix the Alembic migration tomorrow at 2pm	Creates a task with a due date
Plan my day	Builds a realistic plan from your calendar and tasks
Start a work session	Begins tracking focused time
Stop the session	Ends it and summarises what you did

Memory

Remember that I prefer concise technical explanations	Stores it permanently
Remember that I prefer local models whenever possible	Stores a preference it will act on
What do you know about me?	Reads back what it has stored

Understanding and research

Research the best local wake-word engine for Windows	Starts a full research session
Explain this	Explains the document you have open
What's left?	Lists what remains in the project you are in
How much did I work this week?	Measured time, then its interpretation

Looking at your screen

What's on my screen?	Reads the screen and tells you
What does this error say?	Reads the error and explains it
Read my screen	Reads the visible text
Take a screenshot	Captures and reads it

Using your computer

Open Chrome	Opens the app immediately
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Open YouTube and play *a song name*

Opens and starts it

Type *some text*

Types it where your cursor is

Simple, unambiguous commands like these skip the language model entirely so they happen instantly.

SECTION 7

Letting Ultron see your screen

Ultron can look at your screen and read what is on it - useful for errors, dialogs, and anything easier to show than to describe.

How to use it

Just ask, in the chat or by voice:

- *What's on my screen?*
- *What does this error say?*
- *Read my screen*
- *Take a screenshot*

Ultron captures the screen once, reads the text on it, and answers using what it saw.

The rules it follows

Only when asked	Nothing is captured unless you ask in that message. Ultron never watches continuously.
One frame	A single still image. Not a video, not a stream.
Nothing saved	The image is read and discarded. Nothing is written to disk.
Never leaves	Reading is done by Windows on this machine. No image is ever uploaded.
Recording is separate	Asking Ultron to look is not recording. Recording is a different feature, off by default, and shows a red dot whenever it is on.

If it says it cannot see

Screen reading needs two extra pieces. Install them once:

```
pip install -e "[screen]"
```

If something is missing, Ultron says exactly what and how to fix it, rather than failing quietly.

Turning it off

Set `enabled: false` under `vision` in `config/default.yaml`. Ultron will then decline to look, and say so.

SECTION 8

Connecting the real backend

The interface is finished and waiting. Connecting it is a matter of pointing it at your API.

The one file that matters

`frontend/config.js`. Nothing needs rebuilding when you change it - reload the page.

mode	auto checks for a backend and uses it if it answers, otherwise demo data. Live always uses it. mock always uses demo data.
apiBaseUrl	Where the backend is. Empty means the same address serving the page.
apiPrefix	The path your API sits under. Default <code>/api</code> .
eventsPath	Live updates. Set to <code>null</code> if you have none yet.
auth	bearer token, session cookie, or none for a single-user machine.

What to build first

`docs/API.md` in the frontend folder is the full contract: every address, what it sends, what it must send back. Six endpoints are enough to stop being a demo:

- `/system/health` - so Ultron can find the backend
- `/me` and `/settings` - so it knows who you are
- `/chat/stream` and `/conversations` - the conversation
- `/memories` - so it remembers

Anything you have not built yet can return *not found*. That screen shows a message with a Retry button and everything else keeps working.

Where it stands

Ready to connect	Chat, memory, tasks, automations, knowledge, settings - the services already exist in the desktop app.
Not built yet	Agents, research, projects, goals, insights, activity, notifications, integrations and search have no backend yet, and stay on demo data.
Still missing	The desktop app has no web API at all yet. That layer has to be written before anything can be live.

SECTION 9

Your data

Where it lives

On this machine. Conversations, memories, tasks and notes are in a database inside the project's data folder. In demo mode everything is in your browser instead, and clearing site data resets it.

Controls

Settings > Privacy:

Export my data	Downloads a copy as a file.
Reset demo data	Puts the sample content back. Only affects demo mode.
Delete all data	Removes your tasks, memories, documents, conversations, research, automations and history from this device. This cannot be undone, and it really deletes - nothing is hidden or archived.

Principles

- Local first. Nothing leaves this machine unless you connect something that sends it.
- Recording is off until you ask for it, and visible whenever it is on.
- Ultron does not report usage anywhere.
- Deleting means deleting.

SECTION 10

When something is wrong

What you see	What it means	What to do
<i>Demo data - no backend connected</i>	No backend answered, so sample data is being used.	Expected until your API is running. Start it, then reload.
A screen shows an error with Retry	That one request failed.	Press Retry. If it keeps failing, that endpoint is probably not built yet.
<i>Cannot reach Ultron</i> in chat	The backend stopped mid-reply.	Check it is still running, then Retry.
<i>Session expired</i>	Your sign-in is no longer valid.	Sign in again.
Ultron says it cannot see the screen	Screen reading is not installed or is switched off.	Run <code>pip install -e "[screen]"</code> and check <code>vision.enabled</code> .
Nothing loads at all	The page was opened as a file rather than through the server.	Use <code>http://127.0.0.1:5173</code> - opening <code>index.html</code> directly will not work.
Voice does nothing	Microphone or voice setup.	Check Settings, Voice, and that the browser has microphone permission.

APPENDIX

Where things are

<code>frontend/</code>	The interface. Everything needed to run it.
<code>frontend/config.js</code>	Backend address, sign-in, feature switches.
<code>frontend/serve.py</code>	The small server used to run it locally.
<code>frontend/docs/API.md</code>	What the backend must provide.
<code>frontend/docs/PAGES.md</code>	How to change or add a screen.

maira/	The Windows desktop application.
maira/modules/vision/	Screen reading.
config/default.yaml	Desktop app settings.
data/	Your database and files.
Docs/	This manual, the Product Vision and the Build Roadmap.

Ultron is built to make you think in goals rather than software. If a screen ever makes you think about software instead, that is a fault in the screen.